The Risk of Enucleation after Proton Beam Irradiation of Uveal Melanoma

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Abstract: Enucleation after proton beam irradiation of uveal melanomas occurred in 64 (6.4%) of 994 eyes with a median follow-up time of 2.7 years. The median time between irradiation and enucleation in the 64 enucleated eyes was 13 months. The probability of retaining the eye was 95 and 90%, 2 and 5 years postirradiation, respectively. Three percent of eyes were enucleated during posttreatment year 1, and the yearly rate was 1% by the fourth year. No patient had enucleation later than 5½ years posttreatment. The complication most likely to result in enucleation was neovascular glaucoma although this was frequently managed without enucleation. Other common reasons for enucleation were documented or suspected tumor growth and complete retinal detachment with associated loss of vision. The leading risk factors for enucleation were anterior tumor margin involving the ciliary body, tumor height greater than 8 mm, and proximity of the tumor to the fovea. Based on the presence or absence of these factors, 5-year eye retention rates were 99, 92, and 76% for low-, moderate-, and high-risk groups, respectively. Thus, the probability of eye retention after proton beam irradiation is high even among those at greatest risk of enucleation.
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Patients diagnosed with melanoma of the uveal tract face the difficult decision of choosing between enucleation and one of the radiation modalities currently available. Many patients prefer radiation therapy wishing to retain their eyes even when the prognosis for good vision after treatment is poor. Previous reports document the relative success of both radioactive plaque and charged particle (proton and helium ion) radiation therapies in achieving local tumor control.[1–6] Studies also show that a large proportion of eyes retain useful vision after treatment, and tumor characteristics predictive of posttreatment vision have been delineated.[7–9] Charged-particle irradiation offers theoretic advantages over some other methods including: better localization and greater uniformity of dose—properties which should increase the proportion of tumors that can be treated without serious complications necessitating enucleation. We examine enucleation as an outcome in a large series of patients with uveal melanoma treated by proton beam irradiation at the Harvard Cyclotron Laboratory.

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PATIENTS AND METHODS

STUDY POPULATION

The study population consisted of 1006 consecutive patients (1007 eyes) with melanomas of the uveal tract treated at the Harvard Cyclotron between July 1975 and